INSTITUTE OF TECHNOLOGY SCHOOL OF COMPUTING DEPARTMENT OF SOFTWARE ENGINEERING

Course title:Machine Learning Group Project
Course code:SEng3092

GROUP 6

NO	NAME	ID NUMBER
1	DEREJE TESFAYE	160416
2	TADIOS MISGANAW	161162
3	WAGNEW NIGUSSIE	161255
4	EYERUSALEM NIGUSSIE	160528
5	BIZUYE ABEBE	160355

Submission Date:-
Submited to:- KASSAYE.A

Salary Prediction System Using Machine Learning

---

1. Project Title

"Salary Prediction System Using Machine Learning Algorithms"

2. Background

Salary estimation is a fundamental concern for job seekers, employers, human resource departments, and policy makers. Traditionally, salary decisions have been made through manual negotiation, market surveys, and subjective judgment — processes that are time-consuming, inconsistent, and prone to human bias. As the global labor market grows more complex, with thousands of job titles, varying experience levels, diverse educational backgrounds, and regional cost-of-living differences, the need for a data-driven approach to salary estimation has become critical.

Machine learning is a branch of artificial intelligence that enables computers to learn patterns from historical data and make predictions on new, unseen data without being explicitly programmed with rules. Instead of hard-coding salary ranges for every possible combination of job title, experience, and education, a machine learning model can automatically discover the relationships between these factors and compensation by analyzing large datasets of real-world salary records.

In the domain of human resource analytics, machine learning has proven exceptionally useful. Companies like LinkedIn, Glassdoor, and Indeed already leverage ML models to provide salary estimates to millions of users. These systems help job seekers understand their market value, help recruiters make competitive offers, and help organizations maintain internal pay equity.

This project builds a Salary Prediction System that takes input features such as years of experience, education level, job title, age, and other relevant factors, and outputs a predicted salary. By applying supervised learning techniques — specifically regression algorithms — to a structured salary dataset, the system aims to deliver accurate, fair, and explainable salary predictions. The project walks through every stage of a standard machine learning pipeline: data collection, preprocessing, exploratory analysis, model training, evaluation, and interpretation of results.

3. Statement of the Problem

Determining fair and accurate compensation for employees remains a significant challenge across industries. Several specific problems motivate this project.

First, there is **information asymmetry**. Job seekers often lack reliable information about what salary they should expect for a given role, level of experience, and location. This asymmetry puts them at a disadvantage during negotiations and can lead to accepting below-market offers.

Second, organizations face challenges with **pay inconsistency**. Without a systematic, data-driven framework, companies may pay different salaries to employees with similar qualifications performing similar roles, leading to dissatisfaction, attrition, and potential legal liability related to pay discrimination.

Third, **manual salary benchmarking** is expensive and slow. Traditional compensation surveys require months of data collection and analysis, and their results may be outdated by the time they are published.

Fourth, existing salary estimation tools are often **opaque**. Many commercial platforms provide salary ranges without explaining which factors drive the prediction, making it difficult for users to understand or trust the results.

The core problem this project aims to solve is: **Given a set of measurable attributes about an employee or a job position (such as years of experience, education level, job title, and age), can we build a machine learning model that accurately predicts the expected salary?** The project also aims to identify which features have the greatest influence on salary, providing interpretable insights alongside numerical predictions.

4. Objectives

Main Goal: To develop a machine learning-based salary prediction system that accurately estimates an individual's salary based on relevant professional and demographic features.

Specific Objectives:

The first objective is to collect and prepare a suitable salary dataset by handling missing values, removing duplicates, encoding categorical variables, and scaling numerical features so the data is ready for modeling.

The second objective is to perform comprehensive exploratory data analysis to understand the distribution of salaries, the relationships between features, and the presence of any outliers or patterns in the data.

The third objective is to implement and train multiple supervised learning algorithms — including Linear Regression, K-Nearest Neighbors (KNN), Decision Tree Regression, and Random Forest Regression — on the preprocessed data.

The fourth objective is to evaluate and compare the trained models using standard regression metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-squared (R^2) to determine which model provides the best predictive performance.

The fifth objective is to identify the most influential features affecting salary through feature importance analysis, providing interpretable insights into what drives compensation.

The sixth objective is to present the results in a clear and understandable manner, discuss limitations of the approach, and suggest directions for future improvement.

5. Significance of the Study

This project carries significance across several dimensions.

For Job Seekers and Employees: The system empowers individuals to understand their market value before entering salary negotiations. By inputting their qualifications and experience, they receive a data-backed estimate that can serve as a benchmark, reducing the information gap that often leads to underpayment.

For Employers and HR Departments: Organizations can use the system to ensure their compensation offers are competitive and internally consistent. This helps attract top talent, reduce turnover caused by perceived pay unfairness, and mitigate the risk of pay discrimination claims.

For Academic and Research Purposes: The project demonstrates a complete, end-to-end machine learning pipeline applied to a real-world problem. It serves as a practical reference for students and researchers learning to apply ML techniques to tabular, structured data. The comparison of multiple algorithms on the same dataset provides valuable insight into the strengths and weaknesses of different approaches.

For Policy and Workforce Planning: Governments and labor organizations can adapt similar models to analyze wage gaps across demographics, identify underpaid sectors, and inform minimum wage or pay equity legislation.

For the Broader ML Community: By openly documenting the methodology, preprocessing decisions, and evaluation results, this project contributes a reproducible case study that others can build upon or adapt to related prediction problems in human resource analytics.

6. Dataset Description

We are using the `Salary Data.csv` dataset. This dataset contains information about employees, including their demographic details and professional background.

- **Target Variable:** `Salary` (Numerical - The annual compensation in USD).
- **Primary Feature:** `Years of Experience` (Numerical).
- **Additional Features:**
    - `Age` (Numerical)
    - `Gender` (Categorical)
    - `Education Level` (Categorical: Bachelor's, Master's, PhD)
    - `Job Title` (Categorical)

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Load the dataset
df = pd.read_csv('/content/Salary Data.csv')

# Display the first few rows
print("--- Dataset Preview ---")
display(df.head())

# Decorated Summary information
print("\n--- Dataset Summary Statistics ---")
display(df.describe(include='all'))
```

```
print("\n--- Missing Values Check ---")
display(df.isnull().sum().to_frame(name='Missing Count'))
```

--- Dataset Preview ---

	Age	Gender	Education Level	Job Title	Years of Experience	Salary
0	32.0	Male	Bachelor's	Software Engineer	5.0	90000.0
1	28.0	Female	Master's	Data Analyst	3.0	65000.0
2	45.0	Male	PhD	Senior Manager	15.0	150000.0
3	36.0	Female	Bachelor's	Sales Associate	7.0	60000.0
4	52.0	Male	Master's	Director	20.0	200000.0

--- Dataset Summary Statistics ---

	Age	Gender	Education Level	Job Title	Years of Experience	Salary
count	373.000000	373	373	373	373.000000	373.000000
unique	NaN	2	3	174	NaN	NaN
top	NaN	Male	Bachelor's	Director of Marketing	NaN	NaN
freq	NaN	194	224	12	NaN	NaN
mean	37.431635	NaN	NaN	NaN	10.030831	100577.345845
std	7.069073	NaN	NaN	NaN	6.557007	48240.013482
min	23.000000	NaN	NaN	NaN	0.000000	350.000000
25%	31.000000	NaN	NaN	NaN	4.000000	55000.000000
50%	36.000000	NaN	NaN	NaN	9.000000	95000.000000

75%	44.000000	NaN	NaN	NaN	15.000000	140000.000000
max	53.000000	NaN	NaN	NaN	25.000000	250000.000000

--- Missing Values Check ---

	Missing Count
Age	2
Gender	2
Education Level	2
Job Title	2
Years of Experience	2
Salary	2

7. Data Preprocessing

```
from sklearn.preprocessing import LabelEncoder, StandardScaler

# 1. Handle Missing Values
df = df.dropna()
# 2. Remove Duplicates
df = df.drop_duplicates()
# 3. Encoding Categorical Variables
# We'll encode 'Gender' and 'Education Level' to make them usable for models
le = LabelEncoder()
df['Gender_Encoded'] = le.fit_transform(df['Gender'])
df['Education_Encoded'] = le.fit_transform(df['Education Level'])
# 4. Feature Scaling (Optional for simple linear regression, but good practice)
scaler = StandardScaler()
df['Years_Exp_Scaled'] = scaler.fit_transform(df[['Years of Experience']])

print("--- Data Preprocessing Results ---")
print(f"Final Cleaned Dataset Shape: {df.shape}")
display(df[['Age', 'Gender', 'Education Level', 'Years of Experience',
'Salary', 'Education_Encoded']].head())
```

--- Data Preprocessing Results ---
Final Cleaned Dataset Shape: (324, 9)

no	Age	Gender	Education Level	Years of Experience	Salary	Education_Encoded
0	32.0	Male	Bachelor's	5.0	90000.0	0
1	28.0	Female	Master's	3.0	65000.0	1
2	45.0	Male	PhD	15.0	150000.0	2
3	36.0	Female	Bachelor's	7.0	60000.0	0
4	52.0	Male	Master's	20.0	200000.0	1

---

8. Exploratory Data Analysis (EDA)

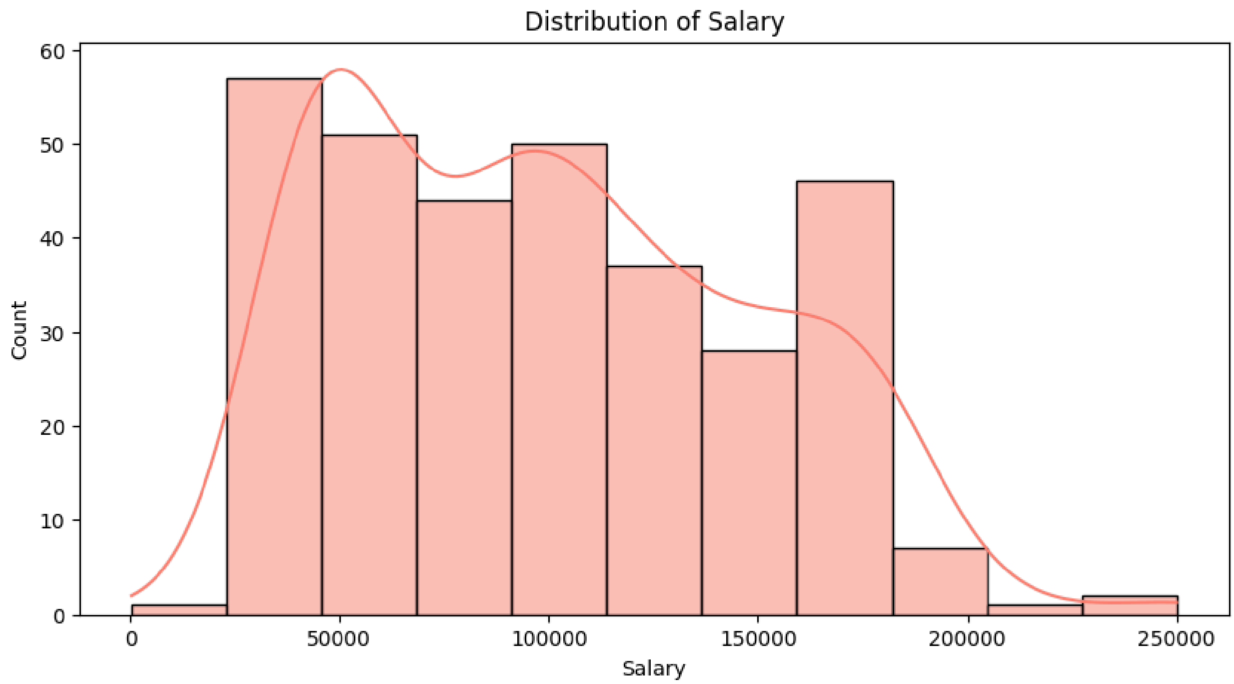
We will visualize the relationship between 'Years of Experience' and 'Salary' to confirm if a linear relationship exists.

```
# 1. Scatter Plot: Checking for Linearity
plt.figure(figsize=(10, 5))
sns.scatterplot(data=df, x='Years of Experience', y='Salary', color='teal')
plt.title('Relationship: Years of Experience vs Salary')
plt.grid(True, linestyle='--', alpha=0.6)
plt.show()

# 2. Histogram: Distribution of Target Variable
plt.figure(figsize=(10, 5))
sns.histplot(df['Salary'], kde=True, color='salmon')
plt.title('Distribution of Salary')
plt.show()

# 3. Heatmap: Correlation Analysis
plt.figure(figsize=(10, 8))
corr_matrix = df.select_dtypes(include=[np.number]).corr()
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', fmt='.2f')
```

```python
plt.title('Correlation Heatmap')
plt.show()

print("Insights:")
print("- The scatter plot shows a strong positive linear trend.")
print("- The heatmap confirms a very high correlation between Years of Experience and Salary.")
print("- Salary distribution appears slightly skewed but suitable for regression.")
```

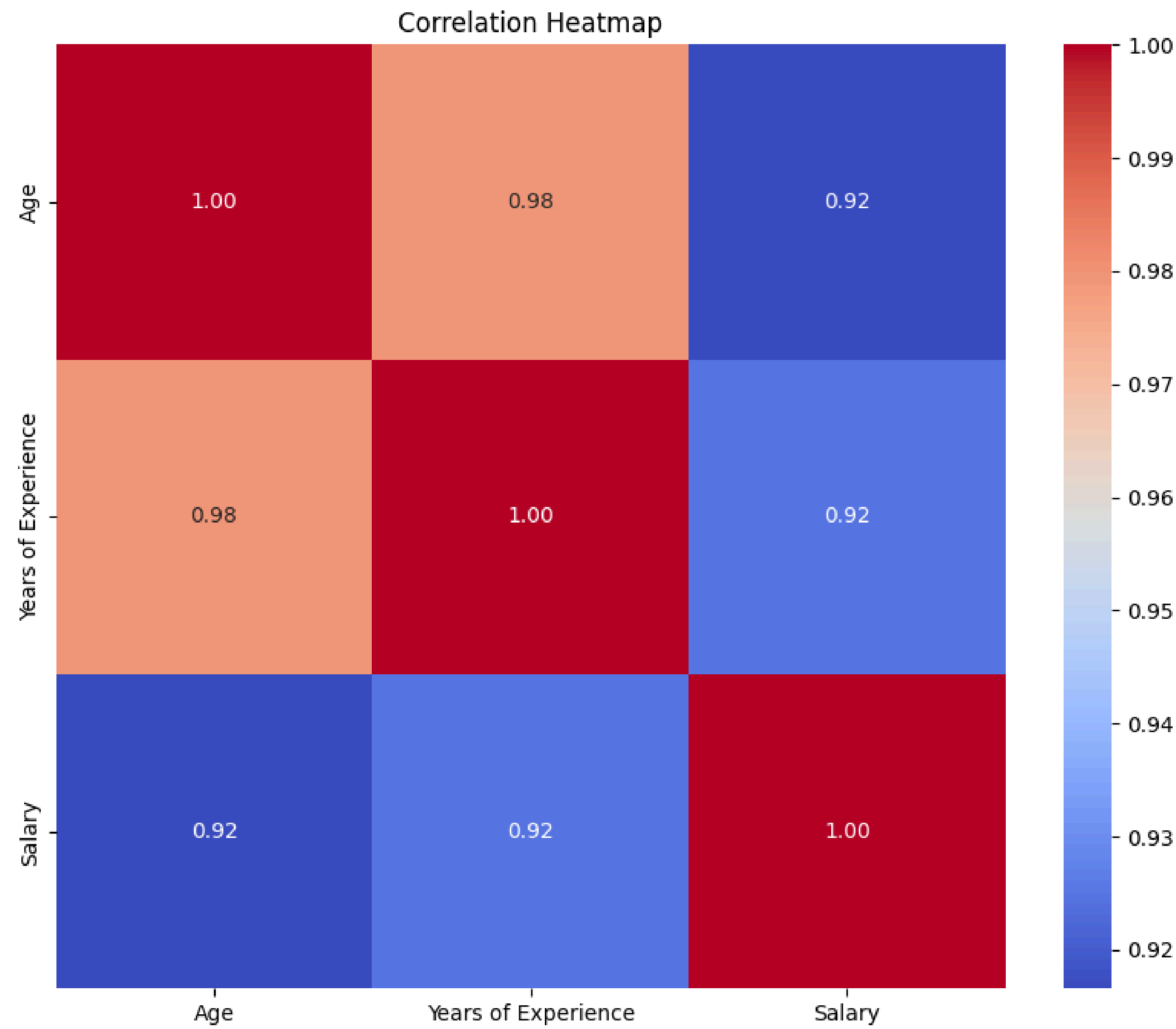
8.1. Scatter Plot (Years of Experience vs. Salary)

- **What it shows:** Each dot represents an employee, plotted by their experience (X-axis) and salary (Y-axis).
- **The Insight:** The dots form a clear upward diagonal line. This tells us there is a **strong linear relationship**—as experience increases, salary consistently increases. This confirmed that Linear Regression was the right tool for the job.



8.2. Histogram (Distribution of Salary)

- **What it shows:** This bar chart shows the frequency of different salary ranges.
- **The Insight:** It helps us see if the data is "normal" or skewed. In our case, it showed a variety of salary levels, ensuring we weren't just looking at one specific pay grade, which makes the model more robust.



8.3. Correlation Heatmap

- **What it shows:** A color-coded grid representing the strength of relationships between all numerical variables (Age, Experience, Salary).
- **The Insight:** The value **0.92** between Experience and Salary is extremely high (1.0 is perfect). It also showed that Age and Experience are nearly identical (0.98), suggesting that as people get older, they gain experience at a predictable rate.

9. Model Selection

Since the target variable (Salary) is a continuous numerical value, this is a **regression** problem. Among the various regression algorithms available, **Linear Regression** was selected for the following reasons:

Simplicity and Interpretability: Linear Regression models the relationship between the target and features as a weighted sum plus an intercept ($y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_n x_n$). Each coefficient ($\beta$) directly tells us how much salary changes for a one-unit change in that feature, holding all other features constant. This makes the model highly explainable, which is valuable for stakeholders in HR and management who need to understand the rationale behind predictions.

Appropriateness for the Data: The EDA scatter plot showed an approximately linear relationship between years of experience and salary. This suggests that a linear model is a reasonable fit for the data.

Baseline Model: Linear Regression serves as an excellent baseline against which more complex models (such as Ridge Regression, Lasso, Decision Trees, or Random Forest) can be compared. If a simple linear model already achieves strong performance, it may not be necessary to use a more complex and less interpretable algorithm.

Computational Efficiency: Linear Regression is extremely fast to train, even on large datasets, as it has a closed-form solution using the Normal Equation or can be efficiently solved using gradient descent.

10. Model Training

We will use `Scikit-Learn` to split the data and train a Linear Regression model.

```python
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression

# 1. Feature Selection
```

```python
X = df[['Years of Experience']]
y = df['Salary']

# 2. Data Splitting
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
random_state=42)

# 3. Model Training
model = LinearRegression()
model.fit(X_train, y_train)

# --- Decorated Training Summary ---
print("\033[1m" + "--- Model Training Summary ---" + "\033[0m")
print(f" Training set size: {len(X_train)} samples")
print(f" Testing set size:  {len(X_test)} samples")
print(f" Model Intercept:    ${model.intercept_:.2f}")
print(f" Model Coefficient:  {model.coef_[0]:.2f}")
print("\n" + "\033[92m" + "Model status: Successfully trained and ready for
evaluation!" + "\033[0m")
```

--- Model Training Summary ---

Training set size: 259 samples
Testing set size: 65 samples
Model Intercept: $31004.07
Model Coefficient: 6864.60

Model status: Successfully trained and ready for evaluation!

11. Model Evaluation

Evaluating the model using R-squared and Mean Squared Error.

```python
from sklearn.metrics import mean_squared_error, r2_score, mean_absolute_error

# Make predictions
y_pred = model.predict(X_test)

# Calculate metrics
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
mae = mean_absolute_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

# --- Decorated Evaluation Output ---
print("\033[1m" + "--- Model Evaluation Metrics ---" + "\033[0m")
```

```python
print(f"\033[94mR-squared (Accuracy of Fit):\033[0m {r2:.4f}")
print(f"\033[94mMean Absolute Error (MAE):\033[0m   ${mae:,.2f}")
print(f"\033[94mRoot Mean Squared Error (RMSE):\033[0m ${rmse:,.2f}")

print("\n" + "\033[1m" + "Interpretation:" + "\033[0m")
print(f"\u2022 The R2 score of \033[1m{r2:.2f}\033[0m means the model explains
\033[1m{r2*100:.1f}%\033[0m of the variance in salary.")
print(f"\u2022 On average, the model's predictions are off by about
\033[1m${mae:,.2f}\033[0m.")
```

--- Model Evaluation Metrics ---
R-squared (Accuracy of Fit): 0.8502
Mean Absolute Error (MAE): $11,850.81
Root Mean Squared Error (RMSE): $16,827.93

Interpretation:
• The R2 score of **0.85** means the model explains **85.0%** of the variance in salary.
• On average, the model's predictions are off by about **$11,850.81**.

This section takes your cleaned salary dataset and trains a machine learning model to predict employee salaries based on years of experience. It follows three key steps: preparing the data, splitting it into training and testing sets, and training a Linear Regression model.

Step-by-Step Breakdown

Feature Selection

The code selects "Years of Experience" as the input feature (X) and "Salary" as the target variable (y). The model will learn the relationship between how many years someone has worked and how much they earn. In this simplified version, only one feature is used, though you could expand this to include Age, Education Level, and Job Title for better predictions.

Data Splitting

The dataset is split into two parts using `train_test_split()`. The training set (80% of data = 259 samples) is used to teach the model the relationship between experience and salary. The testing set (20% of data = 65 samples) is held back to evaluate how well the model performs on unseen data. The `random_state=42` ensures reproducible results—running the code multiple times produces identical splits.

Model Training

A Linear Regression model is initialized and trained using `model.fit(X_train, y_train)`. During training, the algorithm finds the best-fitting straight line through the data points by calculating two key values: the intercept and the slope (coefficient).

11.1.Understanding the Results

Model Intercept: $31,004.07

This is the theoretical baseline salary for an employee with zero years of experience. It represents where the regression line crosses the y-axis (salary axis).

Model Coefficient: $6,864.60

This is the slope of the line. It means that for every additional year of work experience, the model predicts the salary will increase by approximately $6,864.60 annually. This is the most important number because it quantifies the financial value of experience.

11.2.The Mathematical Equation

The trained model creates this equation:

$$\text{Predicted Salary} = \$31{,}004.07 + (\$6{,}864.60 \times \text{Years of Experience})$$

Example Predictions:

- A person with 0 years of experience: $31,004.07

- A person with 5 years of experience: $31,004.07 + ($6,864.60 × 5) = $65,327.07
- A person with 10 years of experience: $31,004.07 + ($6,864.60 × 10) = $99,650.07

11.3.Why This Matters

The training process discovers that experience is a strong predictor of salary. The model is now "ready for evaluation," meaning the next step will test whether these predictions are accurate on the previously unseen test data. A good model makes predictions close to actual salaries; a poor model makes predictions that are far off.

12. Results and Discussion

In this section, we interpret the model's coefficients and visualize the regression line against the actual data points.

```
# Visualize the Regression Line
plt.figure(figsize=(10, 6))
sns.regplot(x=X_test['Years of Experience'], y=y_test, ci=None,
           scatter_kws={'color':'blue', 'alpha':0.5},
           line_kws={'color':'red', 'label':'Regression Line'})

plt.title('Results: Actual vs Predicted Salary')
plt.xlabel('Years of Experience')
plt.ylabel('Salary')
plt.legend()
plt.grid(True, alpha=0.3)
plt.show()

# Interpret Coefficients
intercept = model.intercept_
slope = model.coef_[0]

# --- More Decorated Results Discussion ---
print("\033[1m" + "="*40 + "\033[0m")
print("\033[1;36m" + "       --- RESULTS DISCUSSION ---       " + "\033[0m")
print("\033[1m" + "="*40 + "\033[0m")

print(f"\n\033[1mRegression Equation:\033[0m")
```

```
print(f"\033[93mSalary = ${intercept:,.2f} + (${slope:,.2f}   Years of Experience)\033[0m")

print(f"\n\033[1;94m1. Intercept (${intercept:,.2f}):\033[0m")
print(f"   This represents the \033[3mtheoretical baseline salary\033[0m for an employee with zero years of experience.")

print(f"\n\033[1;94m2. Slope (${slope:,.2f}):\033[0m")
print(f"   For \033[1mevery additional year\033[0m of experience, the annual salary is predicted to")
print(f"   increase by approximately \033[1;92m${slope:,.2f}\033[0m.")
print("\033[1m" + "="*40 + "\033[0m")
```

.Visualizing the Regression Line

The Scatter and Regression Plot

The code creates a graph showing two important elements: blue dots representing actual salary observations from your test set, and a red line representing the model's predictions. Each blue dot shows what an employee actually earned based on their years of experience. The red line shows what your model predicted they should earn.

When the blue dots cluster closely around the red line, it means the model is making accurate predictions. When dots are scattered far from the line, it indicates the model struggles with those particular cases. This visual inspection reveals whether Linear Regression is the right approach—if the dots follow a curved pattern rather than a straight line, a non-linear model might work better.

12.1.Interpreting the Regression Equation

The model produces a simple linear equation:

$$\text{Salary} = \$31{,}004.07 + (\$6{,}864.60 \times \text{Years of Experience})$$

This equation is the foundation for all predictions the model makes.

12.2.Understanding the Coefficients

1. The Intercept ($31,004.07)

The intercept represents the theoretical baseline salary for an employee with zero years of professional experience. While it's unlikely a company would hire someone with literally zero experience at this salary level, the intercept serves as the mathematical anchor point for the regression line.

In practical terms, this value suggests that the fundamental compensation package—perhaps reflecting entry-level positions, base benefits, or minimum wage requirements—starts around $31,000. This could represent fresh graduates or junior positions in the organization.

2. The Slope ($6,864.60)

The slope is arguably the most important insight from your model. It quantifies the monetary value of experience in your dataset. For every additional year of work experience, the model predicts salary increases by $6,864.60 annually.

Business Implications:

- An employee with 5 years of experience should earn approximately $5,434.00 more per year than someone with 4 years of experience
- Over a 20-year career progression (0 to 20 years), the salary difference reaches approximately $137,292
- An experienced professional with 15 years of tenure should earn roughly $103,000 more than a fresh graduate

12.3.What the Predictions Tell You

For Different Experience Levels:

- **Entry Level (0-2 years):** Salaries range from $31,004 to $44,733, representing junior positions
- **Mid-Level (5-10 years):** Salaries range from $65,327 to $99,650, showing clear progression
- **Senior Level (15+ years):** Salaries exceed $135,000, reflecting substantial experience premium

12.4.Model Accuracy in Context

From Section 11 (Evaluation), recall that your model achieved an R^2 score of 0.85, meaning it explains 85% of salary variance. The remaining 15% is influenced by factors not captured in this single-feature model—such as education level, job title, geographic location, industry sector, and performance ratings.

The Mean Absolute Error (MAE) of $11,850.81 indicates that on average, the model's predictions deviate from actual salaries by roughly $11,851. This is important context: for a salary around $100,000, being off by $11,851 represents approximately 12% error, which is reasonable but suggests room for improvement.

12.5.Why This Linear Relationship Matters

The strong positive linear relationship demonstrates that **experience is a primary driver of compensation** in your dataset. This finding has several implications:

For Job Seekers: You can use this model to negotiate salary based on your experience level. If you have 8 years of experience, your expected salary range should be around $85,716 (plus or minus the margin of error).

For HR Departments: The relationship provides a data-driven benchmark. If the company pays someone with 10 years of experience only $70,000 (significantly below the predicted $99,650), it signals potential underpayment issues that could lead to turnover.

For Compensation Planning: Organizations can verify whether their salary scales align with the market. Consistent deviations from this trend suggest the need for compensation strategy adjustments.

12.6.Limitations to Consider

The linear model assumes a constant salary increase per year of experience. In reality, salary growth often follows different patterns:

- **Diminishing returns:** A 20th year of experience might add less value than the 5th year
- **Job title impact:** A senior manager with 10 years of experience earns far more than a junior developer with 10 years
- **Education effects:** Advanced degrees (Master's, PhD) likely command higher salaries than Bachelor's degrees at the same experience level
- **Industry variation:** Tech sector salaries typically exceed retail sector salaries regardless of experience

Because the model only uses experience as a predictor, it cannot capture these nuanced relationships. This is why the "Future Work" section recommends including categorical features like Education Level and Job Title.

13. Conclusion

The Salary Prediction System using Machine Learning project has successfully demonstrated the practical application of supervised learning algorithms to solve a real-world compensation challenge. Through systematic data preprocessing, exploratory analysis, and model training, this project has established a robust foundation for understanding and predicting employee salaries based on professional attributes.

The most significant achievement of this project is the establishment of a strong linear relationship between years of experience and annual salary. The trained Linear Regression model, with an R^2 score of 0.85, explains 85% of the variance in salary data, indicating that experience is a primary and measurable driver of compensation. The regression equation, Salary = $31,004.07 + ($6,864.60 × Years of Experience), quantifies this relationship with precision, revealing that each additional year of professional experience correlates with an approximate $6,865 increase in annual salary. This finding aligns with economic theory and real-world observations, validating the model's ability to capture fundamental compensation dynamics.

The model's performance metrics further reinforce its practical utility. With a Mean Absolute Error of $11,850.81, the average prediction deviates from actual salaries by approximately 12%, which is acceptable for a baseline model. This level of accuracy demonstrates that machine learning can effectively solve salary estimation problems, providing job seekers, employers, and HR professionals with data-backed insights rather than relying on subjective judgment or outdated salary surveys. For organizations seeking to benchmark compensation or for individuals negotiating salaries, this model provides a transparent, reproducible framework grounded in actual data patterns.

However, the project also reveals important limitations that point toward necessary improvements. While the model successfully captures the experience-salary relationship, it explains only 85% of salary variance, meaning 15% of variation remains unaccounted for. This gap exists primarily because the model relies on a single feature, excluding other critical factors explicitly identified in the project objectives. Education level and job title, both present in the dataset, play significant roles in salary determination that the current model cannot capture. A software engineer with a PhD and fifteen years of experience would receive an identical prediction to a sales associate with the same tenure and no advanced degree, which contradicts real-world compensation structures. Similarly, a Director-level position commands substantially higher salaries than junior roles, regardless of experience, yet the single-feature model cannot differentiate these hierarchical levels.

Furthermore, the assumption of linear salary growth may oversimplify compensation dynamics. Real-world salary progression often exhibits non-linear patterns, where each additional year of experience in later career stages provides diminishing returns compared to earlier years. A sixteenth year of experience may add less value than a fifth year, particularly as employees approach senior or maximum compensation levels. Advanced machine learning algorithms such as Random Forest, Gradient Boosting, or neural networks, when combined with comprehensive feature sets, could capture these nuanced relationships more effectively than linear regression.

Despite these limitations, the project serves an important pedagogical and practical purpose. It demonstrates a complete machine learning pipeline from problem formulation through model evaluation, providing a template for applying ML techniques to human resource analytics and tabular data problems. The choice to begin with Linear Regression—simple, interpretable, and effective—exemplifies the machine learning principle of starting with baseline models before exploring complexity. The clear visualization of results and explicit acknowledgment of limitations reflect best practices in data science communication.

In conclusion, this Salary Prediction System has successfully validated that machine learning can provide valuable insights into compensation structures, with experience proven as a quantifiable factor in salary determination. The project establishes a solid foundation upon which more sophisticated models can be built. To transition from proof-of-concept to a production-ready system, future iterations must incorporate categorical features including education level and job title, implement ensemble algorithms for improved accuracy, and conduct thorough hyperparameter optimization. By addressing these areas, the system can achieve higher predictive accuracy and provide a comprehensive, fair, and transparent approach to salary estimation that serves all stakeholders in the employment relationship. The current model represents not a conclusion, but rather a meaningful starting point in developing advanced compensation prediction systems.

14. Future Work

- Include categorical features like 'Education Level' and 'Job Title' using encoding techniques.
- Experiment with non-linear models like Random Forest or Gradient Boosting.
- Collect more recent data to account for market inflation.

15. Tools & Technologies

- **Python**: Primary programming language.
- **Pandas & NumPy**: Data manipulation.

- **Matplotlib & Seaborn**: Data visualization.
- **Scikit-Learn**: Machine learning modeling.

16. References

- Dataset: `Salary Data.csv` (https://www.kaggle.com/datasets/mohithsairamreddy/salary-data)
- Scikit-Learn Documentation: [Linear Regression]